

Worksheet — Number Systems (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions in your notebook. Show every step of your work for the subjective questions; choose the single best option for MCQs. Answers without working will not receive full marks.
Total time: 45 minutes.

Q1. Which of the following is the smallest whole number? [1 mark]

- (A) -1
- (B) 0
- (C) 1
- (D) There is no smallest

Q2. Which inclusion chain is correct? [1 mark]

- (A) $W \subset N \subset Z \subset Q$
- (B) $N \subset W \subset Z \subset Q$
- (C) $Q \subset Z \subset W \subset N$
- (D) $Z \subset W \subset Q \subset N$

Q3. Every integer is a rational number. [1 mark]

- (A) True
- (B) False

Q4. The reciprocal of $-5/8$ is _____. [1 mark]

Q5. Express $-24/36$ in standard (lowest) form: [2 marks]

- (A) $-2/3$
- (B) $-24/36$
- (C) $-4/6$
- (D) $2/-3$

Q6. Evaluate: $(-2/3) + (5/6) - (1/2)$. Show all steps. [3 marks]

Q7. Find three rational numbers between $\frac{1}{4}$ and $\frac{1}{3}$. [3 marks]

Q8. Without performing long division, decide whether $\frac{13}{40}$ and $\frac{7}{15}$ have terminating decimal expansions. Justify each decision. [4 marks]

Q9. Express $0.\overline{6}$ and $0.\overline{2}7\overline{7}$ (bar over the digits shown) as fractions in lowest terms. Show the algebraic manipulation in each case. [5 marks]

Q10. (a) Verify the distributive property of multiplication over addition for rational numbers using $a = \frac{3}{4}$, $b = \frac{1}{3}$, $c = \frac{5}{6}$. (b) Then state and verify which of the four 'addition' properties (closure, commutative, associative, additive identity) hold for rational numbers, giving a one-line example for each. [9 marks]

Answer Key

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Q1. Which of the following is the smallest whole number?

Answer: 0

Whole numbers $W = \{0, 1, 2, \dots\}$; the smallest is 0.

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Q2. Which inclusion chain is correct?

Answer: $N \subset W \subset Z \subset Q$

Naturals $\subset$ Wholes $\subset$ Integers $\subset$ Rationals.

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Q3. Every integer is a rational number.

Answer: True

Any integer n equals $n/1$, which is rational.

TRUE_FALSE · EASY · 10-NIOS-MATH-NS-04

Q4. The reciprocal of $-5/8$ is _____.

Answer: $-8/5$

Reciprocal of a/b is b/a , preserving sign: reciprocal of $-5/8$ is $-8/5$.

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Q5. Express $-24/36$ in standard (lowest) form:

Answer: $-2/3$

HCF(24, 36) = 12; divide both by 12 $\rightarrow -2/3$ with positive denominator.

MCQ · MEDIUM · 10-NIOS-MATH-NS-04

Q6. Evaluate: $(-2/3) + (5/6) - (1/2)$. Show all steps.

Answer: LCM of 3, 6, 2 is 6. Rewrite: $-2/3 = -4/6$, $5/6 = 5/6$, $1/2 = 3/6$. Then $-4/6 + 5/6 - 3/6 = (-4 + 5 - 3)/6 = -2/6 = -1/3$.

Use a common denominator (LCM 6), then combine numerators and reduce.

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Q7. Find three rational numbers between $\frac{1}{4}$ and $\frac{1}{3}$.

Answer: Bring to a common denominator with extra room: $\frac{1}{4} = \frac{9}{36}$, $\frac{1}{3} = \frac{12}{36}$. Three rationals strictly between are $\frac{10}{36} = \frac{5}{18}$, $\frac{11}{36}$ and (using 144 as denominator) $\frac{41}{144}$ = also between. So $\frac{5}{18}$, $\frac{11}{36}$ and $\frac{41}{144}$ are three rationals strictly between $\frac{1}{4}$ and $\frac{1}{3}$. (Other valid answers exist; the key is to refine the denominator until at least three integers lie strictly between the rewritten numerators.)

Rationals are dense — choose any common denominator d for which $d \times (\frac{1}{3}) - d \times (\frac{1}{4}) \geq 4$ to leave at least three integer numerators between them.

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Q8. Without performing long division, decide whether $\frac{13}{40}$ and $\frac{7}{15}$ have terminating decimal expansions. Justify each decision.

Answer: $\frac{13}{40}$: $40 = 2^3 \times 5$. Lowest-terms denominator has only the primes 2 and 5, so the decimal terminates. (Indeed $\frac{13}{40} = 0.325$.) $\frac{7}{15}$: $15 = 3 \times 5$. The factor 3 is not 2 or 5, so the decimal is non-terminating recurring. (Indeed $\frac{7}{15} = 0.466\ldots = 0.4\overline{6}$.)

Apply the rule: p/q (lowest terms) terminates iff $q = 2^m \times 5^n$.

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Q9. Express $0.\overline{6}$ and $0.\overline{27}$ (bar over the digits shown) as fractions in lowest terms. Show the algebraic manipulation in each case.

Answer: Part 1 — $0.\overline{6}$: Let $x = 0.666\ldots$. Multiply by 10: $10x = 6.666\ldots$. Subtract: $10x - x = 6 \rightarrow 9x = 6 \rightarrow x = \frac{6}{9}$. Reduce by HCF 3: $x = \frac{2}{3}$. So $0.\overline{6} = \frac{2}{3}$. Part 2 — $0.\overline{27}$ (bar over 27): Let $x = 0.272727\ldots$. Multiply by 100 (because 2 digits repeat): $100x = 27.2727\ldots$. Subtract: $100x - x = 27 \rightarrow 99x = 27 \rightarrow x = \frac{27}{99}$. Reduce by HCF 9: $x = \frac{3}{11}$. So $0.\overline{27} = \frac{3}{11}$.

For an n -digit repeating block, multiply by 10^n then subtract the original to eliminate the recurring tail.

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Q10. (a) Verify the distributive property of multiplication over addition for rational numbers using $a = \frac{3}{4}$, $b = \frac{1}{3}$, $c = \frac{5}{6}$. (b) Then state and verify which of the four 'addition' properties (closure, commutative, associative, additive identity) hold for rational numbers, giving a one-line example for each.

Answer: (a) Distributive: $a(b + c) = ab + ac$. LHS: $(\frac{3}{4}) \times (\frac{1}{3} + \frac{5}{6}) = (\frac{3}{4}) \times (\frac{2}{6} + \frac{5}{6}) = (\frac{3}{4}) \times (\frac{7}{6}) = \frac{21}{24} = \frac{7}{8}$. RHS: $(\frac{3}{4})(\frac{1}{3}) + (\frac{3}{4})(\frac{5}{6}) = \frac{3}{12} + \frac{15}{24} = \frac{6}{24} + \frac{15}{24} = \frac{21}{24} = \frac{7}{8}$. LHS = RHS, so the property holds. (b) Closure: For any $p/q, r/s \in \mathbb{Q}$, $p/q + r/s \in \mathbb{Q}$. Example: $\frac{2}{5} + \frac{3}{7} = \frac{14}{35} + \frac{15}{35} = \frac{29}{35} \in \mathbb{Q}$. Commutative: $a + b = b + a$. Example: $\frac{1}{2} + \frac{1}{3} = \frac{5}{6} = \frac{1}{3} + \frac{1}{2}$. Associative: $(a + b) + c = a + (b + c)$. Example with $a = \frac{1}{2}$, $b = \frac{1}{3}$, $c = \frac{1}{4}$: $(\frac{1}{2} + \frac{1}{3}) + \frac{1}{4} = \frac{5}{6} + \frac{1}{4} = \frac{13}{12}$; $\frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) = \frac{1}{2} + \frac{7}{12} = \frac{13}{12}$. Equal. Additive identity: there exists $0 \in \mathbb{Q}$ with $a + 0 = a$. Example: $\frac{7}{9} + 0 = \frac{7}{9}$. All four properties hold.

Verifying a property algebraically with one numeric example is enough at this level.

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